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(c) What is Animation? Explain following terms related with animantion

(i) In-betweening

(ii) Procedural Animation

(iii) Key framing

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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(IT Engg. Branch)

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COMPUTER GRAPHICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Answer any two part from each question. Each question carries equal marks.*

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1. (a) Explain any two techniques to achieve color picture in CRT Monitor, with their principle and application area for which they best suited.

(b) What is Transformation? Explain following transformation and give the homogenous matrix.

(i) Translations

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(ii) Rotation

(iii) Scaling

(c) What is Clipping? Explain Hodgeman-Sutherland clipping algorithm for polygon.

2. (a) What are splines? Explain meaning of following terms for the splines.

(i) Interpolation

(ii) Approximation

(iii) Convex hull

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(iv) Convex graph

(v) Various continuity conditions

(b) Discuss characteristics of B-Spline curves. Generate equations for blending function for the cubic splines considering $d = 3$.

(c) What are the properties of Bezier curve? Give the polynomial equation for Bezier curve.

3. (a) Explain and compare the perspective projection and parallel projection with example.

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(b) What do you mean by Hidden Surface Removal? What are the different techniques. Explain A-buffer method for Hidden Surface Removal?

(c) Explain in brief about Scan line algorithm for removal of hidden surface.

4. (a) Explain gouraud shading technique. Explain deficiency of this method, how it is rectified using phong shading tech.

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(b) What is Texture? Discuss various texture mapping technique.

(c) Discuss illumination model for diffused and specular reflection.

5. (a) Explain different types of fractals also explain how to construct fractals and use of fractals in computer graphics.

(b) Discuss the Ray tracing process with an example.